Shadman Sohel

905-251-2304 | shadman.sohel04@gmail.com | linkedin.com/in/shadman-sohel | shadmansohel.onrender.com

Education

Toronto Metropolitan University (TMU) Expected Graduation: 2027
Bachelor of Computer Engineering *Toronto, Ontario*

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Systems, Software Design
- **GPA:** 3.81/4.33

Experience

Software Developer Intern May 2026 – August 2026
Google *Kitchener, Ontario*

- Improved RAG response quality by **5%** for enterprise customer agents, by engineering a **Kotlin/Java** chunk reranker that prioritizes high-value context within strict LLM token limits.
- Validated sub-**300ms** performance across **500** queries, by migrating **20** agents and **30** datastores into end-to-end **Python** evaluation pipelines.
- Maintained system reliability across **1,000** daily RPCs, by utilizing extensible stubs for query request hedging and establishing real-time monitoring dashboards with automated alerts.
- Isolated false positives across **20** evaluations, by developing an **AI agent skill** that orchestrates tool calls for Sheets comparisons and subagents for insight validation.

Computer Engineering Intern April 2025 – May 2026
Ontario Power Generation (OPG) *Toronto, Ontario*

- Automated **5,000** PDF reactor safety reviews, by engineering a weekly **Python/MySQL** cron pipeline.
- Reduced manual overhead by **78%** and intercepted **100** input errors in 3 months, by building an **LLM**-powered **Python** form input companion to validate field entries.

Software Engineering Intern September 2024 – March 2025
RaiseIt *Toronto, Ontario*

- Delivered **400** legislative items to over **100** government officials, by building a full-stack **React/Node.js** portal.
- Boosted push notification engagement by **15%** across **10** feature rollouts, by engineering **AWS CloudWatch** dashboards to analyze A/B campaign performance.
- Summarized **100+** daily transcripts into MPP profiles, by crafting a **Node.js**, **Lambda** and **MongoDB** pipeline.

Projects

Morsum - Social Media App | *React Native, Spring Boot, PostgreSQL, PyTorch, AWS* Download Link

- Onboarded **125** testers to an AI-powered food social app, built with **React Native** and **Spring Boot**.
- Attained **75%** prediction accuracy on user uploads, by engineering a self-retraining **PyTorch** image classification microservice via **FastAPI** that triggers **EC2** workflows from **S3 Lambda** events.
- Achieved sub-**400ms** query latency, by indexing frequent queries for a **PostgreSQL** database on **AWS ECS**.

AskFYEO - Q & A Chatbot | *Python, SQL, Hugging Face* Website

- Served over **2,000** students via a **Python** chatbot, using Hugging Face to query a **MySQL vector DB**.
- Expanded the dataset by **100** entries in 4 months via a Python feedback loop capturing common user queries.
- Prototyped an **MCP server** to standardize data access within AI agent integration across university domains.

Battle Bites - Multiplayer Game | *ReactJS, NodeJS, MongoDB* Website | Devpost | Source Code

- Delivered a online multiplayer game within 48-hours at **Hack the 6ix** using **React.js**, **Node.js**, and **MongoDB**.
- Achieved **sub-2ms** latency for the game loop by optimizing the **Socket.io** pipeline using batched state updates.
- Diversified gameplay with real-time **AI-generated** power-ups, processing skill files via **S3 Lambda** event.

Technical Skills

Languages: (Proficient) Kotlin, Java, JavaScript/TypeScript, (Intermediate) Python, C++, SQL
Frameworks/Libraries: Spring Boot, ReactJS, NextJS, NodeJS, React Native, FastAPI, Flask
Developer Tools: Google Cloud Platform, AWS, Postman, Git, Docker, Power BI